ICT GCE A Level 2023 - Paper 3

Duration: 2h

Cameroon GCE Board

General Certificate of Education Board - Advanced Level Examination. Information & Communication Technologies, Subject code 0796, Paper 3 Group 1. Examination date: June 2023. Duration: two hours.

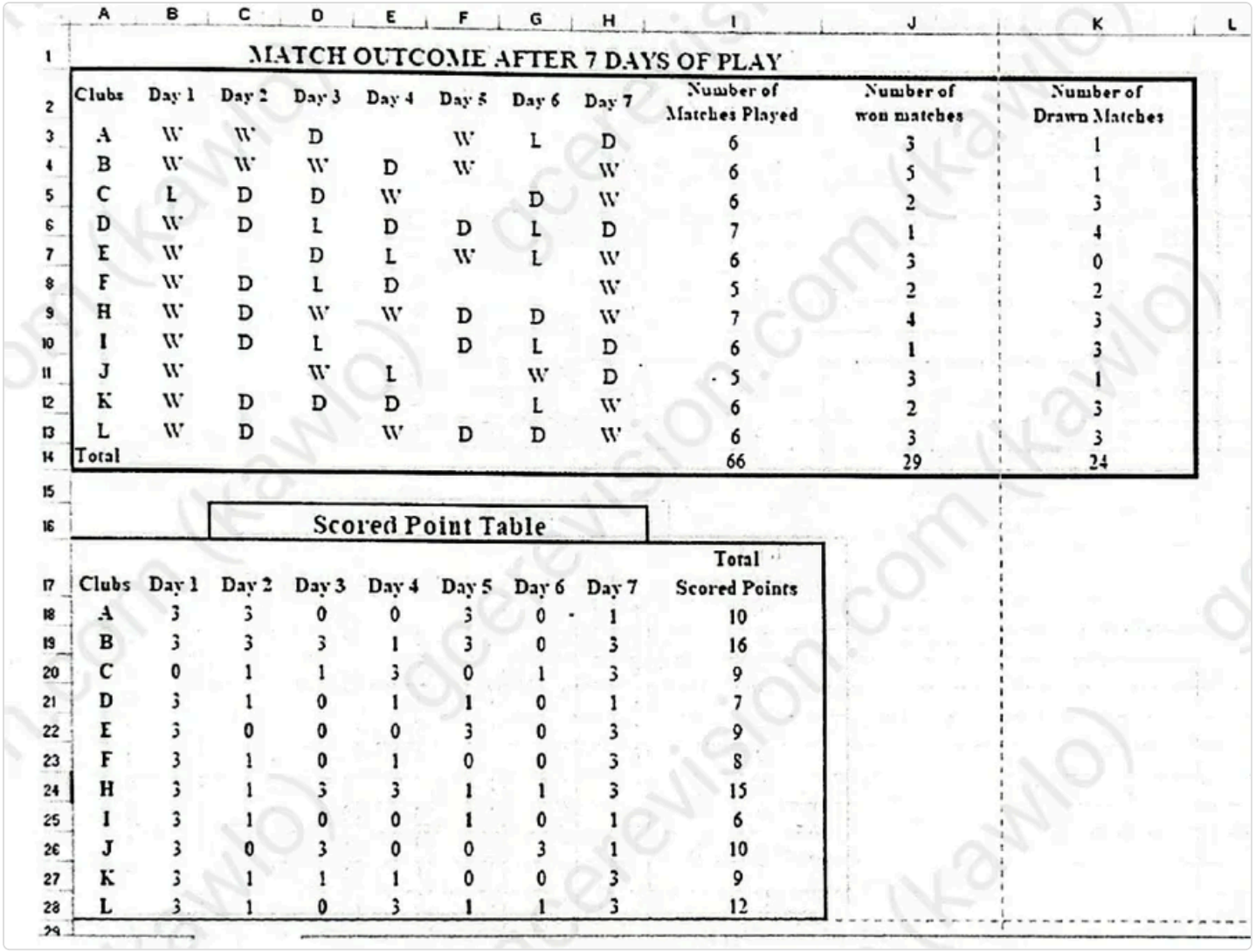
Enter the information required in the shaded boxes. Do not write in pencil. You are reminded of the necessity for good English and orderly presentation in your answer. Your results must be recorded in the spaces provided in this question booklet. Candidates must allow for themselves enough time to complete and check their work where these are required. The supervisor will guide you on how to save your files but usually you will be expected to save all your work in one folder named *Candidate folder* that will be created in your desktop. When an imperative programming language is required to write program code, either Standard [ISO] Pascal or [ANSI] C programming language may be used. You should ensure all your work is collected and printed before leaving the examination room. All printed work should be inserted in this booklet and well stapled. Carefully handle all equipment put at your disposal to avoid accidents that may lead to a disaster or system failure. Report any case of system or device malfunction to your supervisor.

Answer ALL TASKS.

The paper has three tasks: Task 1 (Presentation, 17 marks), Task 2 (Spreadsheet, 25 marks) and Task 3 (Programming, 7 marks).

Exercise 2 (25 marks)

TASK 2 (Spreadsheet: 25 marks)



MATCH OUTCOME AFTER 7 DAYS OF PLAY

Clubs	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Number of Matches Played	Number of won matches	Number of Drawn Matches
A	W	W	D		W	L	D	6	3	1
B	W	W	W	D	W		W	6	5	1
C	L	D	D	W		D	W	6	2	3
D	W	D	L	D	D	L	D	7	1	4
E	W		D	L	W	L	W	6	3	0
F	W	D	L	D			W	5	2	2
H	W	D	W	W	D	D	W	7	4	3
I	W	D	L		D	L	D	6	1	3
J	W		W	L		W	D	5	3	1
K	W	D	D	D		L	W	6	2	3
L	W	D		W	D	D	W	6	3	3
Total								66	29	24

Scored Point Table

Clubs	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Total Scored Points
A	3	3	0	0	3	0	1	10
B	3	3	3	1	3	0	3	16
C	0	1	1	3	0	1	3	9
D	3	1	0	1	1	0	1	7
E	3	0	0	0	3	0	3	9
F	3	1	0	1	0	0	3	8
H	3	1	3	3	1	1	3	15
I	3	1	0	0	1	0	1	6
J	3	0	3	0	0	3	1	10
K	3	1	1	1	0	0	3	9
L	3	1	0	3	1	1	3	12

Data of a football league of eleven clubs (A to L) is displayed in the table. The table shows the outcome of the matches played after 7 playing days. In the first table, a match won is represented by W, drawn by D, and lost by L. The second table shows the scored points per club. The points are gotten by converting the outcome into scored points where a win (W) is 3 points, a draw is 1 point, and a loss is 0 point.

The picture above is the printed spreadsheet *MATCH OUTCOME AFTER 7 DAYS OF PLAY* (title in row 1, headings in row 2, clubs in rows 3 to 13, totals in row 14) followed by the *Scored Point Table* (title in row 16, headings in row 17, clubs in rows 18 to 28). A blank cell means that the club did not play that day. The printed values are recopied below, cell by cell, with the spreadsheet column letters.

Table 1: Match outcome after 7 days of play (as printed)

Row	A (Clubs)	B (Day 1)	C (Day 2)	D (Day 3)	E (Day 4)	F (Day 5)	G (Day 6)	H (Day 7)	I (Number of Matches Played)	J (Number of won matches)	K (Number of Drawn Matches)
3	A	W	W	D		W	L	D	6	3	1
4	B	W	W	W	D	W		W	6	5	1
5	C	L	D	D	W		D	W	6	2	3
6	D	W	D	L	D	D	L	D	7	1	4
7	E	W		D	L	W	L	W	6	3	0

Row	A (Clubs)	B (Day 1)	C (Day 2)	D (Day 3)	E (Day 4)	F (Day 5)	G (Day 6)	H (Day 7)	I (Number of Matches Played)	J (Number of won matches)	K (Number of Drawn Matches)
8	F	W	D	L	D			W	5	2	2
9	H	W	D	W	W	D	D	W	7	4	3
10	I	W	D	L		D	L	D	6	1	3
11	J	W		W	L		W	D	5	3	1
12	K	W	D	D	D		L	W	6	2	3
13	L	W	D		W	D	D	W	6	3	3
14	Total								66	29	24

Table 2: Scored Point Table (as printed)

Row	A (Clubs)	B (Day 1)	C (Day 2)	D (Day 3)	E (Day 4)	F (Day 5)	G (Day 6)	H (Day 7)	I (Total Scored Points)
18	A	3	3	0	0	3	0	1	10
19	B	3	3	3	1	3	0	3	16
20	C	0	1	1	3	0	1	3	9
21	D	3	1	0	1	1	0	1	7
22	E	3	0	0	0	3	0	3	9
23	F	3	1	0	1	0	0	3	8
24	H	3	1	3	3	1	1	3	15
25	I	3	1	0	0	1	0	1	6
26	J	3	0	3	0	0	3	1	10
27	K	3	1	1	1	0	0	3	9
28	L	3	1	0	3	1	1	3	12

1. Open the spreadsheet named *Elite_FBL* found in your candidate folder. It contains the data above. *[1 mark]*

2. Insert a thick outside border in the range A2:K14. (See the second table) *[2 marks]*

3. Merge the cells in the range C16:H16 and increase the font size of the content to 18 and Font bold. *[2 marks]*

4. Wrap the text in cell I17 as shown in the picture above. *[2 marks]*

5. Type a formula in cell I3 to calculate the number of matches played. Copy this formula to compute the same for other clubs. Write down the formula you have in the following cell.

I9: ..... *[4 marks]*

6. Insert a formula in cell J3 to compute the number of matches won. Copy this formula to compute the same for other clubs. Write down the formula you have in the following cell.

J3: ..... *[4 marks]*

7. Insert a formula in cell K3 to compute the number of matches drawn. Copy this formula to compute the same for other clubs. Write down the formula you have in the following cell.

K6: ..... *[2 marks]*

8. Type a conditional formula in B18 that will show the points scored by each club per match given that a win (W) is 3 points, draw (D) is 1 point, and a loss (L) is 0 point. Copy this formula to compute the same for other days in all the clubs. Write down the formula you have in the following cells.

B18: .....

G23: ..... *[4 marks]*

9. Calculate the total scored points scored by each team. Write down the formula you have in the following cells.

I25: ..... *[4 marks]*

10. Print a copy of your work.

Exercise 3 (7 marks)

Task 3 (Programming: 7 marks)

Below is the algorithm to calculate the difference of two integers. You are expected to convert the algorithm into a code in either Pascal or C programming language. Part of the code is given in the table below.

Algorithm

```
Start
    Type an integer, and name it as A.
    Type another integer, and name it as B.
    Calculate: Diff = A - B.
    Print Diff.
Stop.
```

After a successful compilation, use the lines in the table below to copy the missing instructions. Cancel the corresponding section of code not used.

Pascal code	**C code**
Program rectAB;	#include
Var A, B, Diff: integer;	int main () {
Begin	int A, B, Diff;
(seven blank lines for the missing instructions)	*(seven blank lines for the missing instructions)*

Pascal code	C code
End.	return 0; }

Save your code using the name *Rect_Task* in your candidate folder and print a copy. *[7 marks]*

Want to check your answers?

The full correction of this paper, with the method explained step by step, is available on EduKora: 15,000 FCFA per child until the exam, 12,000 FCFA for each additional child in the same family.

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